

Alam Bushara

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Education

University of Colorado Boulder

Boulder, CO

B.S. Computer Science, Minor in Business Analytics

May 2025

Coursework: Algorithms, Data Structures, Database Systems, Computer Systems, Software Development, Data Science, Artificial Intelligence, Cyber-Security

Experience

Boeing Defense, Space & Security

Huntsville, AL

Quality Data Engineer

November 2025 – Present

- Developed Python data pipelines and transformations in Palantir Foundry code repositories, integrating enterprise production and quality data to power automated reporting, non-conformance analysis, and reliability monitoring.
- Built Foundry Ontologies unifying engineering, manufacturing, and quality data to support the accelerated production ramp of advanced seeker systems.
- Shipped operational applications and AI-powered workflows using Foundry AIP and LLM capabilities to automate quality investigations and accelerate root cause analysis.
- Enforced data governance, role-based access controls, and security policies across sensitive engineering systems, ensuring audit compliance and integrity of production-critical data assets.

SB Energy

Denver, CO

Intern, Performance Engineering

May 2025 – October 2025

- Constructed Python analytics pipelines monitoring utility-scale solar performance, processing high-frequency time-series data to automate KPI reporting and surface degradation patterns across distributed plant systems.
- Built data ingestion and transformation systems processing large-scale drone-captured thermal and visual datasets, integrating outputs into Git-managed production workflows for anomaly detection and output optimization.

National Security Agency

Boulder, CO

Intern, Software

August 2024 – May 2025

- Engineered an open-source Software-Defined Radio (SDR) toolkit for digital signal and protocol analysis, enhancing security assessments and RF threat detection in an Agile environment.
- Developed PyQt5 real-time signal visualization tools featuring FFT analysis, multithreaded processing, and modular architecture, collaborating through Git version control and code reviews.
- Reduced training time by 10%, saving over \$400,000 per recruit by optimizing core software workflows and streamlining onboarding.

Groove Beverages Inc.

Boulder, CO

Co-Founder & CTO

May 2023 – June 2025

- Co-founded a beverage startup, contributing to product formulation and company direction from inception.
- Designed, built, and deployed the company's public-facing website (HTML, CSS, JavaScript), owning the full stack from development through production hosting, domain infrastructure, and ongoing maintenance.

Projects

Raumdeuter | Football Prediction Engine

Python, JavaScript, React, StatsBomb, Vitest

github.com/aabu4537/raumdeuter • raumdeuter.vercel.app

- Built a Python ETL pipeline ingesting StatsBomb open-data match events (2022 World Cup, 16 matches) into per-90 percentile player ratings, validated by 22 pytest unit tests over the aggregation logic.
- Engineered a DOM-free JavaScript prediction engine (Dixon-Coles adjusted Poisson, tunable on/off-ball α blend) with an exact analytical bracket solver replacing Monte Carlo simulation, recomputing championship odds client-side in real time and covered by an 11-test property-based Vitest suite.
- Validated the engine against real historical results and six pre-committed 2026 knockout predictions (each frozen before kickoff): 0.184 Brier score vs. a 0.25 coin-flip baseline, with 5 of 6 favorite calls correct.

Skills

Languages: Python, C++, JavaScript, TypeScript, SQL, Bash/Shell

Data & Analytics: Palantir Foundry, Foundry AIP, Ontologies, ETL, Data Modeling, Pandas, NumPy, scikit-learn, XGBoost, Tableau, Power BI

Web & Backend: Node.js, React.js, Next.js, FastAPI, REST APIs, HTML, CSS

Databases: PostgreSQL, Microsoft SQL Server, MySQL, MongoDB, Oracle

Infrastructure: Git, Docker, Linux, CI/CD, Redis

Methods: Statistical Process Control (SPC), Root Cause & Corrective Action (RCCA), AS9100, Agile